

AI vs. Cyber Threats: Measuring Insights, Accuracy, and Actionability in CVE analysis



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Research Question

How do AI models like DeepSeek and ChatGPT perform when querying outside of their knowledge base?

What makes a good response?

- **Insightfulness:** How detailed or useful are the responses?
- **Verifiability:** Can the information be confirmed by reputable sources?
- **Actionability:** Does the output provide clear, relevant next steps or mitigations?

Methodology

- **CVEs:** *Common Vulnerabilities and Exposures*
 - A standardized framework for cybersecurity professionals to discuss vulnerabilities.
- **Experiment Setup:**
 - Compare outputs from **ChatGPT o1** and **DeepSeek V3** using a **standardized prompt** and a **CVE ID**.
 - Use CVE IDs from **2020-21** (known in model training) and **2024-25** (unknown in foundational knowledge).

You are a cybersecurity expert.

Query: {CVE_ID}

Given the specified CVE-ID, please provide detailed answers to the following questions: How can an attacker exploit this vulnerability? Provide a step-by-step description. What are the recommended mitigation strategies for this vulnerability?

Evaluation Metrics

Metrics	Method	Purpose
Insightfulness	G-Eval, Answer Relevancy	Measure depth, uniqueness, and usefulness of responses
Verifiability	DAG, ProveRAG	Ensure claims are backed by reputable sources and logical dependencies
Relevance	Prompt Alignment	Check if answers strictly follow prompt instructions
Hallucination	NLI Score	Detects incorrect or misleading information

Implementation

- **Use LLM-as-a-judge:** Use AI to evaluate outputs on a clear scale of 1 to 5 for each metric.
- **Reference & Date Check:** Require citations or references to recent sources to ensure up-to-date information

Why It Matters?

- **AI limitations:** GPT models lack real-time updates, leading to inaccurate CVE advice and security risks.
- **Trustworthiness:** Establishing a systematic way to measure AI reliability.
- **Human-AI collaboration:** Helping professionals focus on solutions rather than manual verification.

Sources

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